

# SmartCampus — Intelligent Campus Route Finder

Single-Source Shortest Path Optimization using Dijkstra's Algorithm and Min-Binary Heap

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## 1. Introduction

Navigating large modern educational campuses with numerous interconnected academic blocks, research labs, libraries, food courts, and sports facilities presents a daily spatial optimization challenge. **SmartCampus** models the campus physical layout as a weighted graph  $G = (V, E)$  and executes Dijkstra's Algorithm with a Min-Binary Heap to compute shortest walkable paths in  $O((V + E) \log V)$  time.

## 2. Problem Statement

Given campus graph  $G = (V, E)$  with  $|V|$  vertices and  $|E|$  walkways with non-negative distances  $w(u, v) \geq 0$ , find an ordered path  $P = \langle s = v_0, v_1, v_2, \dots, v_k = t \rangle$  that minimizes total physical distance  $\sum w(v_{i-1}, v_i)$  between starting vertex  $s$  and destination  $t$ .

## 3. Objectives

- Represent campus infrastructure as an Adjacency List graph.
- Implement Dijkstra's Algorithm using a custom Binary Min-Heap Priority Queue.
- Provide an interactive step-by-step visual simulator for academic evaluation.
- Benchmark algorithmic performance against BFS, A\* Search, and Bellman-Ford.
- Deliver an interactive Graph Editor allowing arbitrary graph modifications.

## 4. Graph Representation $G = (V, E)$

**Vertices (V):** Main Block, Library, CS & AI Block, Central Cafeteria, Auditorium, Science Labs, Sports Arena, Hostel Residence, Examination Block, Health Center, Research Hub, Amphitheatre.

| Edge ID    | Source (u) | Target (v)        | Distance (w) | Pathway Type   |
|------------|------------|-------------------|--------------|----------------|
| e-main-lib | Main Block | Knowledge Library | 120 m        | Covered Avenue |

| Edge ID    | Source (u)        | Target (v)        | Distance (w) | Pathway Type      |
|------------|-------------------|-------------------|--------------|-------------------|
| e-lib-cs   | Knowledge Library | CS & AI Block     | 180 m        | Academic Corridor |
| e-main-cs  | Main Block        | CS & AI Block     | 340 m        | Central Spine     |
| e-main-caf | Main Block        | Central Cafeteria | 200 m        | Canopy Walkway    |
| e-caf-cs   | Central Cafeteria | CS & AI Block     | 100 m        | Plaza Connector   |

## 5. Formal Pseudocode

```
ALGORITHM Dijkstra(Graph G, Source s, Destination t):
1.  FOR each vertex v in G.V:
2.      dist[v] ← INFINITY
3.      prev[v] ← NULL
4.  dist[s] ← 0
5.
6.  PQ ← MinPriorityQueue()
7.  PQ.insert(s, 0)
8.
9.  WHILE PQ is not empty:
10.     u ← PQ.extractMin()
11.     IF u = t: BREAK // Early stopping
12.
13.     FOR each neighbor v in G.Adj[u]:
14.         alt ← dist[u] + weight(u, v)
15.         IF alt < dist[v]: // Edge Relaxation
16.             dist[v] ← alt
17.             prev[v] ← u
18.             PQ.insert(v, alt)
19.
20. RETURN ReconstructPath(prev, s, t), dist[t]
```

## 6. Asymptotic Complexity Analysis

| Operation        | Adjacency Matrix | Binary Min-Heap (SmartCampus) | Fibonacci Heap    |
|------------------|------------------|-------------------------------|-------------------|
| Time Complexity  | $O(V^2)$         | $O((V + E) \log V)$           | $O(E + V \log V)$ |
| Space Complexity | $O(V^2)$         | $O(V + E)$                    | $O(V + E)$        |

## 7. Test Suite Assertions

| Test ID | Scenario               | Source → Target             | Expected    | Actual      | Status |
|---------|------------------------|-----------------------------|-------------|-------------|--------|
| T1      | Direct Edge            | Main Block → Library        | 120m        | 120m        | PASS   |
| T2      | Multi-Hop Corridor     | Library → CS Block          | 180m        | 180m        | PASS   |
| T3      | West Quad Edge         | Main Block → Auditorium     | 160m        | 160m        | PASS   |
| T4      | Self-Loop (Same node)  | Main Block → Main Block     | 0m          | 0m          | PASS   |
| T5      | Optimal Path Selection | Main Block → CS Block       | 300m        | 300m        | PASS   |
| T6      | Disconnected Island    | Main Block → Isolated Annex | UNREACHABLE | UNREACHABLE | PASS   |

## 8. Algorithmic Comparison: Dijkstra vs BFS

In the canonical scenario (*Main Block → CS Block*), BFS selects the direct 1-hop edge of **340m** because it only minimizes edge count. In contrast, **Dijkstra's Algorithm** explores the 2-hop route via Library (120m + 180m = **300m**), discovering the true shortest physical path and saving 40 meters.

## 9. Conclusion

SmartCampus demonstrates the practical power of graph algorithms and greedy optimization. The binary min-heap implementation delivers provably optimal physical routing with  $O((V + E) \log V)$  performance, making complex graph theory transparent and interactive.